

PORT-GENTIL, Gabon

WMO: 645010

Lat: 0.712S

Lon: 8.754E

Elev: 4

StdP: 101.28

Time Zone: 1.00 (E01)

Period: 11-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	22.0	19.0	13.8	23.4	19.7	14.4	23.5	8.3	26.2	7.9	25.9	3.1	180	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	5.4	32.0	27.1	31.2	27.0	31.1	26.9	27.8	30.4	27.5	30.1	27.2	29.8	4.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.8	29.9	26.8	22.4	29.7	26.2	21.7	29.1	89.1	30.9	87.3	30.6	86.3	30.6	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	7.1	6.5	20.6	33.8	0.5	2.3	20.2	35.4	19.9	36.7	19.6	38.0	19.3	39.7		
(5)			18.7	28.8	2.0	0.4	17.3	29.1	16.1	29.3	14.9	29.5	13.4	29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.9	27.9	28.2	28.4	28.2	27.6	25.7	24.4	25.1	26.4	27.0	27.0	27.4	
	DBStd	1.56	0.98	1.08	0.96	0.94	0.93	1.15	1.03	0.81	0.66	0.74	0.93	0.90	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD10.0	6181	556	509	572	544	544	470	446	468	491	528	512	541	
	CDD18.3	3139	298	276	313	294	286	220	187	210	241	270	262	282	
	CDH23.3	30133	3366	3212	3712	3398	3004	1512	773	1142	1944	2529	2576	2964	
	CDH26.7	7739	1053	1129	1386	1188	788	161	34	57	222	438	539	744	
Wind	WSAvg	3.8	3.3	3.2	3.2	3.1	3.6	4.5	4.5	4.7	4.9	4.2	3.6	3.2	
Precipitation	PrecAvg	1905	254	176	266	289	142	2	1	6	28	175	355	235	
	PrecMax	2595	443	435	461	556	390	11	25	28	100	440	669	547	
	PrecMin	1333	35	56	116	44	0	0	0	0	1	20	168	58	
	PrecStd	332	101	82	85	141	111	4	4	8	23	113	128	131	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.8	32.1	32.2	32.2	31.8	29.2	28.2	28.8	29.3	30.2	30.8	31.2	
		MCWB	27.4	27.1	27.2	27.1	27.0	25.3	23.9	24.5	25.1	26.0	26.5	26.7	
	2%	DB	31.1	31.4	32.0	31.8	31.0	28.9	27.5	27.9	29.0	29.8	30.1	30.6	
		MCWB	27.0	27.1	27.1	27.0	26.9	25.1	23.5	23.9	25.0	25.7	26.2	26.6	
	5%	DB	30.8	31.1	31.2	31.1	30.2	28.1	26.9	27.2	28.3	29.1	29.7	30.0	
		MCWB	26.8	27.0	27.0	26.9	26.6	24.6	23.2	23.4	24.5	25.4	26.0	26.3	
	10%	DB	30.1	30.6	31.0	30.8	29.8	27.8	26.1	26.8	28.0	28.8	29.1	29.4	
		MCWB	26.4	26.8	26.9	26.9	26.4	24.4	22.9	23.3	24.3	25.2	25.7	26.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.0	28.2	28.2	28.2	27.7	26.2	24.6	25.0	26.1	26.7	27.1	27.3
			MCDB	30.5	30.6	30.7	30.6	29.8	28.3	27.0	27.3	28.3	29.0	29.2	29.7
2%		WB	27.4	27.7	27.7	27.7	27.2	25.7	24.1	24.4	25.7	26.2	26.6	26.9	
		MCDB	29.9	30.3	30.4	30.2	29.6	27.9	26.5	26.7	27.8	28.6	28.8	29.3	
5%		WB	27.1	27.3	27.4	27.3	26.9	25.2	23.7	24.0	25.2	25.9	26.3	26.6	
		MCDB	29.6	29.9	30.2	29.9	29.3	27.3	26.0	26.3	27.4	28.2	28.4	28.9	
10%		WB	26.7	27.0	27.1	27.0	26.6	24.8	23.3	23.7	24.8	25.6	26.1	26.3	
		MCDB	29.2	29.7	30.0	29.7	28.9	26.9	25.6	25.9	27.0	27.9	28.2	28.6	
Mean Daily Temperature Range	5% DB	MDBR	4.5	4.9	5.4	5.3	4.3	3.8	3.9	3.6	3.8	4.0	4.3	4.3	
		MCDBR	4.9	5.1	5.8	5.8	5.0	4.0	4.3	4.0	4.1	4.3	4.8	4.9	
		MCWBR	2.4	2.4	2.8	2.8	2.4	1.9	1.9	1.8	1.9	2.0	2.3	2.2	
	5% WB	MCDBR	4.7	4.9	5.4	5.4	4.8	3.8	4.1	3.8	4.0	4.2	4.5	4.6	
		MCWBR	2.5	2.5	2.9	2.9	2.5	2.0	1.9	1.9	2.0	2.1	2.5	2.3	
Clear-Sky Solar Irradiance	taub	0.637	0.619	0.564	0.468	0.512	0.607	0.656	0.631	0.515	0.461	0.497	0.612		
	taud	1.855	1.903	2.060	2.374	2.193	1.902	1.786	1.848	2.178	2.364	2.265	1.906		
	Ebn at Noon	726	746	784	842	781	692	664	703	811	866	835	738		
	Edh at Noon	214	207	176	125	144	189	214	208	154	129	141	202		
All-Sky Solar Radiation	RadAvg	4.51	4.57	4.80	4.68	4.36	4.17	4.12	4.16	4.37	4.46	4.12	4.23		
	RadStd	0.16	0.23	0.21	0.24	0.37	0.28	0.20	0.26	0.22	0.24	0.18	0.24		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page